

Quick Check

Lesson 5: Water, Mineral Salts and Enzymes in Digestion

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 5

Wanjiru's blood glucose rose from 4.5 to 8.0 mmol/L after she ate ugali, which is mostly starch. Why must starch be digested before it can raise her blood glucose?

- A** Starch is a very large insoluble molecule and must be broken down into small soluble sugars first
- B** Starch is already glucose, but it needs to be warmed to 37 degrees C first
- C** Starch turns directly into blood once it is swallowed
- D** Saliva only makes the food wet so it can be swallowed, and this is all that is needed

In an investigation, tube A has starch and saliva. Tube B has starch and boiled saliva. Both are kept at about 37 degrees C. Why is tube B included?

- A** To make sure there is more liquid to test with iodine
- B** To show that boiled saliva digests starch faster than fresh saliva
- C** To test whether Benedict's solution can change colour without heating
- D** To show that the changes in tube A are caused by active saliva and not by warmth or time alone

At 10 minutes, a sample from the tube of starch and fresh saliva at 37 degrees C gives a yellow-brown colour with iodine and a brick-red colour with Benedict's solution after heating. What do these results show?

- A** Starch is still present, and the saliva has turned into a reducing sugar
- B** Starch has been used up and a reducing sugar is now present
- C** Starch is still present and no sugar has formed
- D** Starch has increased and Benedict's solution has turned it into protein

A tube of starch and saliva was boiled before the test and stayed blue-black with iodine after 10 minutes. What is the correct explanation?

- A** Boiling turned the enzyme into starch
- B** Boiling made the starch molecules too big to be digested
- C** Boiling used up all the enzyme by making it work too fast
- D** Boiling denatured the enzyme, so it could no longer digest starch

A tube of starch and fresh saliva is placed in ice at 0 degrees C. After 10 minutes it is still blue-black with iodine. What does this tell us about the enzyme?

- A** The enzyme is inactive at this low temperature
- B** The enzyme works fastest at 0 degrees C
- C** The enzyme is not needed to digest starch
- D** The enzyme has been denatured and destroyed for ever by the cold

Water is an inorganic compound found in the starch solution and in saliva. What is its main role in the digestion of starch?

- A** It is the enzyme that breaks starch into sugar
- B** It is the medium in which the enzyme and starch react and in which the digested sugars are carried
- C** It makes the enzyme a protein
- D** It changes the iodine from blue-black to yellow-brown

A group is revising the 'Digestion by enzymes' box in the class model. Which statement about salivary amylase should they write?

- A** It is a protein catalyst that speeds up the digestion of starch to maltose, and it is not used up
- B** It is a lipid that stores energy in the mouth
- C** It is a carbohydrate that is used up as it digests starch
- D** It is a mineral salt that works best at 0 degrees C

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).